Scale-equalizing Pyramid Convolution for object detection

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Abstract

*Feature pyramid has been an efficient method to extract features at different scales. Developments over this method mainly focus on aggregating contextual information at different levels while seldom touching the inter-level correlation in the feature pyramid. Early computer vision methods extracted scale-invariant features by locating the feature extrema in both spatial and scale dimension. Inspired by this, a convolution across the pyramid level is proposed in this study, which is termed pyramid convolution and is a modified 3-D convolution. Stacked pyramid convolutions directly extract 3-D (scale and spatial) features and outperforms other meticulously designed feature fusion modules. Based on the viewpoint of 3-D convolution, an integrated batch normalization that collects statistics from the whole feature pyramid is naturally inserted after the pyramid convolution. Furthermore, we also show that the naive pyramid convolution, together with the design of RetinaNet head, only best applies for extracting features from a Gaussian pyramid, whose properties can hardly be satisfied by a feature pyramid. In order to alleviate this discrepancy, we build a scale-equalizing pyramid convolution (SEPC) that deforms the shared pyramid convolution kernel only at high-level feature maps. Being computationally efficient and compatible with the head design of most single-stage object detectors, the SEPC module brings significant performance improvement ($>$ 4*AP* increase on *MS-COCO2017 dataset*) in state-of-the-art one-stage object detectors, and a light version of SEPC also has $\sim$ 3.5*AP* gain with only around 7% inference time increase. The pyramid convolution also functions well as a stand-alone module in two-stage object detectors and is able to improve the performance by $\sim$ 2*AP*.*

1. Introduction

An object may appear in vastly different scales in natural images and yet should be recognized as the same. The scales can easily vary by more than 1 magnitude in natural images[31], which presents as a challenging task in various computer vision tasks such as object detection. Extensive research has focused on this issue. Multi-scale training [4] is a direct solution to scale changes by letting the network memorize the patterns at different scales. Multi-scale inference [25] shares the same idea with traditional image pyramid methods[24, 28]. However, the image pyramid method is time-consuming since multiple inputs are necessary. Intrinsic feature pyramid[22] in CNNs at different stages provides an efficient alternative to image pyramid. Each level of the downsampled convolutional features corresponds to a specific scale in the original image. However, there exists a semantic gap between each two levels in feature pyramid. To alleviate the discrepancy, different feature fusion strategies have been proposed, including top-down information flow[17, 6], an extra bottom-up information flow path [21, 38, 13], multiple hourglass structures[26, 44], concatenating features from different layers [16, 34, 9], feature

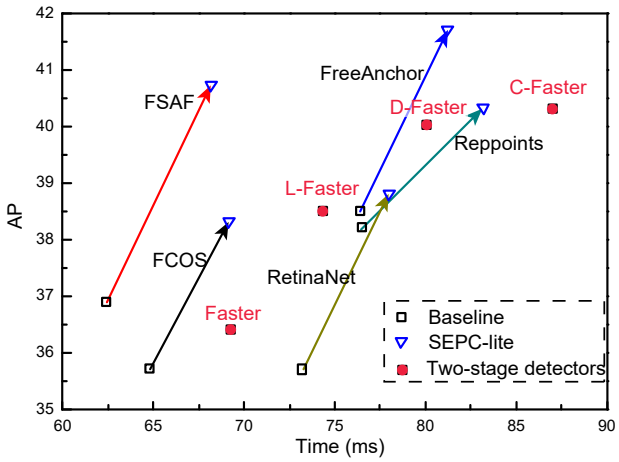

Figure 1: Pyramid convolution performance on various single-stage detectors including RetinaNet[18], FCOS[35], FSAF[45], Reppoints[41], FreeAnchor[43]. Reference points of two-stage detectors such as Faster R-CNN (Faster)[29], Libra Faster R-CNN (L Faster)[27], Cascade Faster R-CNN (C Faster)[1] and Deformable Faster R-CNN (D Faster) [46] are also provided.

	p3	p4	p5	p6	p7
p3	1.00	0.86	0.70	0.00	0.00
p4	0.55	1.00	0.84	-0.01	0.00
p5	0.26	0.50	1.00	-0.01	0.00
p6	0.15	0.29	0.46	1.00	0.48
p7	0.09	0.16	0.24	0.48	1.00

Figure 2: Correlation matrix of feature maps in the feature pyramid of RetinaNet. The upper and lower triangle represents the correlation before and after FPN respectively.

refinements using non-local attention module[27], gradual multi-stage local information fusions [42, 33]. However, the design of feature fusion has become intuitive, and several traits of feature maps of different scales are missing in these feature fusion modules.

For instance, feature maps of neighboring scales should interact the most, which is however not captured in previous methods. A feature pyramid is built by extracting intermediate outputs after each downsample operation of a feature-extracting network (backbone), such as VGG[30], ResNet[12] and Resnext[40]. There are multiple layers of convolution and non-linearities in between each feature maps extracted from the backbone. Therefore, neighboring feature maps in the feature pyramid should correlate more than the remote ones. This is validated in Fig.2, which plots the correlation matrix between each two feature maps extracted from the backbone before and after FPN in RetinaNet. Values close to the diagonal ones are observed larger than remote ones. This is similar to the prior of using spatial convolutions for handling natural image that neighboring pixels on an image correlates stronger than distant pairs. However, this property is not directly captured in previous feature fusion designs.

Moreover, in traditional computer vision methods, a scale space is created and treated as an extra dimension besides the spatial dimension. And a useful feature is identified as a local minimum in both spatial and scale dimensions[20]. Motivated by these, we propose to capture the inter-scale correlations through an explicit convolution in the scale dimension, forming a 3-D convolution in the feature pyramid, which is termed pyramid convolution (PConv).

Further, we also show that this naive design of PConv, together with the head of RetinaNet, a special case of PConv with scale kernel = 1, actually best suits for extracting features from a Gaussian pyramid. A Gaussian pyramid is generated by consecutively blurring an image with a Gaussian kernel followed by a subsampling. The kernel size of the Gaussian blur should be proportional to the subsampling ratio, so as to remove high-frequency noise during subsampling and not big enough to remove much detail. Conducting PConv in this Gaussian pyramid helps extract scale-invariant features.

However, the feature pyramid constructed from a deep backbone network is usually far from a Gaussian pyramid. First, the multiple-layers of convolution in the backbone between two feature pyramid levels make a larger effective Gaussian kernel; Second, the theoretical value of effective Gaussian kernel should vary from pixel to pixel due to the non-linearity operations such as ReLU in obtaining the next pyramid features. As a result, we explore the possibility of relaxing these two discrepancies by devising a scale-equalizing pyramid convolution (SEPC) module. It caters for both the stronger blurring effect and the pixel-specific blurring kernel in the feature pyramid. Using the idea of deformable convolution[46], the kernel size at the bottom pyramid is fixed and deforms as PConv strides in the scale dimension. Such a design enables better extracting scale-invariance features from feature pyramid and only brings a modest computational cost increase since the deformable kernels are only applied to high-level features. Equipped with the final SEPC module, the detection performance boosts for various models. For example, SEPC module is tested to be able to reach as high as 4.3AP increase in state-of-the-art single stage detectors, such as FreeAnchor[43], FSAF[45], Reppoints[41] and FCOS[35], making them even surpasses most two-stage detectors. A light version of SEPC (SEPC-lite) can also reach a performance gain of around 3.5AP with only around 7% increase in computational cost.

This study mainly contributes in the following aspects.

(1). We propose a light-weighted pyramid convolution (PConv) to conduct 3-D convolution inside the feature pyramid to cater for inter-scale correlation.

(2). We also propose a scale-equalizing pyramid convolution (SEPC) to relax the discrepancy between the feature pyramid and the Gaussian pyramid by using a shared convolutional kernel and only deforming it at high-level feature maps.

(3). The module boosts the performance ($\sim$ 3.5AP increase on state-of-the-art single stage object detectors) with negligible inference speed compromise.

2. Related work

2.1. Object detection

Modern object detection architectures are generally divided into one-stage and two-stage ones. Two-stage detection representatives like SPP[11], Fast R-CNN[8], Faster R-CNN[29] first extract region proposals and then classify ob-

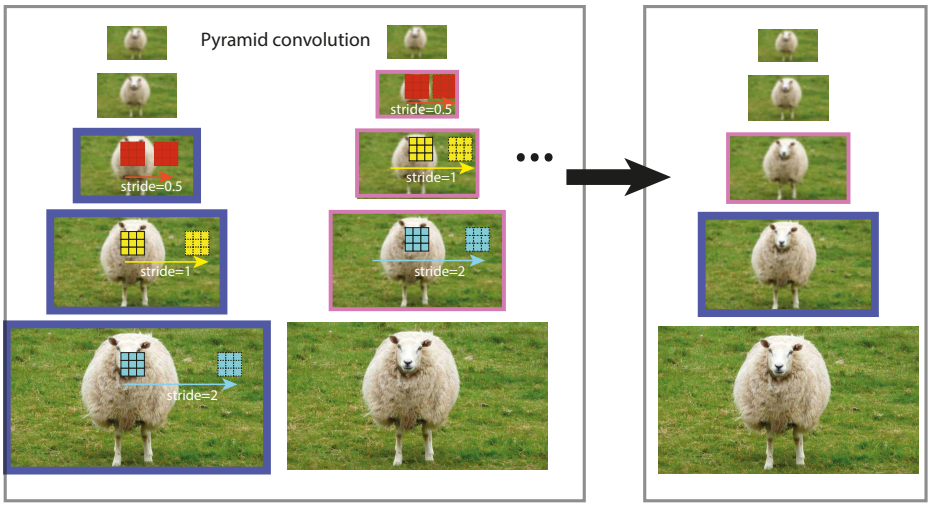

Figure 3: Pyramid convolution as a 3-D convolution. Three convolutional kernels (in red, yellow and cyan) are used for this 3-D convolution. The convolutional stride of each kernel scales as the size of the feature map. Feature maps of the same frame color (e.g. blue and pink) generates the feature map at the right side of the same frame color. The image is only used to show the scale and does not represent feature maps.

jects inside. The scale variance problem is somewhat mitigated in two-stage detectors where object of different sizes are rescaled to be the same size during the ROI pooling process. On the other hand, single-stage object detection[22] directly utilizes the intrinsic sliding-window trait of convolutions to build feature pyramids and directly predict objects based on each pixel. Though having earned advantage in real-time tasks due to its fast inference, single-stage detectors has been lagging behind two-stage ones as for the performance. RetinaNet[18] is a milestone single-stage detector since it boosts detection performance by adopting focal loss and new design of detection head. Following works further accelerated the model and improved its performance simultaneously by viewing object detection as key point localization tasks and thus removing the dependency on multiple anchors at each feature map[41, 35]. But the design of FPN and head remains the same as RetinaNet.

2.2. Feature fusion

In deep networks, low-level features are generally deemed lacking in semantic information but rich in keeping geometric details, which is the opposite for high-level features. Therefore, feature fusion plays a crucial rule in combining both semantic and geometric information. FPN and its contemporary works leveraged high-level feature maps when detecting small objects. The following works further the efficiency of feature fusion from different aspects. As shown in Fig. 4, PA-Net then found out that detecting large objects also needs the assistance of location-sensitive feature maps, and directly created a short path for low-level feature maps. Following the same philosophy, multiple bidirectional information fusion paths were also proposed in[26, 44]. Apart from normal approaches of direct summation, some other methods also adopted concatenation to project all feature maps to a common space followed by a back distribution. Pang et al. [27] furthered the level of feature diffusion by adding a non-local block to fine-tune the combined feature maps.

2.3. Cross-scale correlation

There have also been several other methods considering the cross-scale correlation in both traditional and recent research. Cross-scale difference was calculated to approximate the Laplacian operator in SIFT[23] to extract scale-invariant features. Worrall & Welling[39] also extended group convolution to deep neural networks using dilated convolution. Wang et al. [37] fused feature maps with the neighboring scales to capture inter-scale correlation after all feature maps are transferred to be of the same size as the largest one. In these works, either repeated computations over different transformation of the input image is needed [39] or scale correlation is conducted on a high-resolution feature map[37], both of which incurs undesirable increase in computational resources. In this study, the pyramidal structure of feature maps is maintained when conducting convolution across different scales, which is much more efficient in computation. Actually, the original design of the head structure of RetinaNet and its descendants can also be viewed as a pyramid convolution with scale kernel of 1. Therefore, our design of pyramid convolution is totally

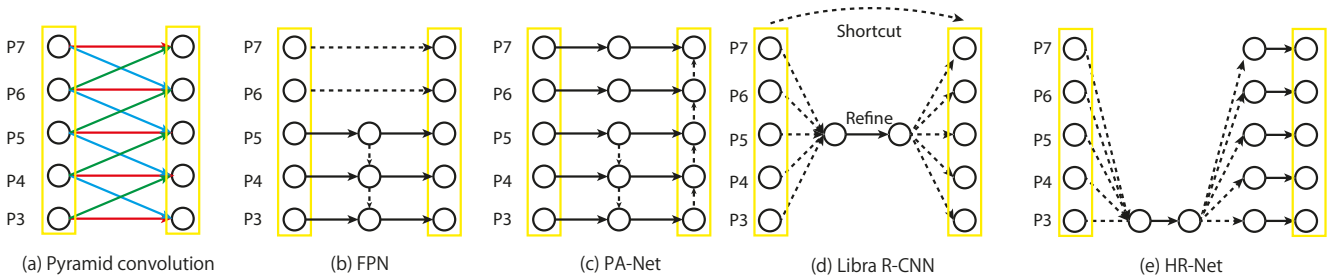

Figure 4: Comparison of pyramid convolution (a) with other feature fusion modules, including (b) FPN[17] (c) PA-Net[21], (d) Libra R-CNN [27] and (e) HR-Net[34]. Each feature map upward has a spatial sized scaled down by two by default. Dotted lines represent interpolation operations, meaning that they can be upsampling, downsampling or shortcut depending on the respective feature map sizes. Each black solid line is an independent convolution, and colored solid lines of the same color are shared convolution operations.

compatible with the state-of-the-art single-stage object detectors with minimal computational cost increase.

3. Pyramid convolution

The pyramid convolution is indeed a 3-D convolution across both scale and spatial dimensions. As shown in Fig. 4a, if we represent the features in each level as a dot, and the pyramid convolution degrades to a normal 1-D convolution in the scale space with kernel size L. It seems that specifying L spatial convolution kernels suffices to conduct pyramid convolution. Nevertheless, as shown in Fig. 3 there is a size mismatch across different pyramid levels. The spatial size is scaled down by 2 by default as the pyramid level goes up by one. In order to accommodate the mismatch, we set different strides for kernels of one pyramid convolution when convolving in different layers. For example, for PConv of scale size =3, the kernel below the current layer should have a stride of 2 while the one above should have a stride of 0.5. Then the output of the PConv is

$$y^l = w^{l+1} *_{s0.5} x^{l+1} + w^l * x^l + w^{l-1} *_{s2} x^{l-1}, \quad (1)$$

where l denotes pyramid level, w is the convolutional weight, x is the input feature map and $*_{s2}$ means a convolution with stride 2. The kernel of stride 0.5 is further replaced by a normal convolution with stride of 1 and a consecutive bilinear upsampling layer. That is,

$$y^l = \text{Upsample}(w^{l+1} * x^{l+1}) + w^l * x^l + w^{l-1} *_{s2} x^{l-1} \quad (2)$$

Similar to conventional convolutions, zero-padding is also used for pyramid convolution. As for the bottom pyramid level ($l = 1$), the last term in Eq. 2 is unnecessary while for the top-most level ($l = L$), the first term is ignored. Despite the 3 convolution operations at each layer, the total FLOPs of pyramid convolution is actually only around 1.5 times as much as the original head (see Appen. 1).

3.1. Pipeline

Apart from the ability of extracting scale-correlated features, pyramid convolution also benefits from its compatibility with the head design of RetinaNet and its descendants. As seen from Fig.5a, RetinaNet head is actually also a pyramid convolution with a scale kernel of one. Therefore, the 4 convolutional heads can be directly replaced by our pyramid convolution module with a scale kernel of 3. The stacked pyramid convolution echos the stacked convolutions modules in 3-D deep networks[36], so as to gradually increase correlation distance without much computational burden.

However, each pyramid convolution still brings some additional computation. As an alternative, the 4 PConv modules are shared by both classification and localization branch, forming a combined head structure as shown in Fig. 5b. In order to cater for the difference in classification task and localization task, an extra normal convolution are also added after the shared 4 PConv modules. It can be calculated that this design has even less FLOPs than the original RetinaNet head (see Append. 1).

3.2. Integrated Batch normalization in the head

In this study, we also retrieve batch normalization in detection head. A single BN follows the pyramid convolution by collecting statistics from all feature maps inside the feature pyramid, instead of from a single layer. This design comes naturally as we view PConv as a 3-D convolution. Since the statistics are collected from all the feature maps inside the pyramid, the variance becomes smaller, especially for high-level features having small feature map sizes. This enables us to train BN in the head even in a small batch size ~ 4 and achieve better performance.

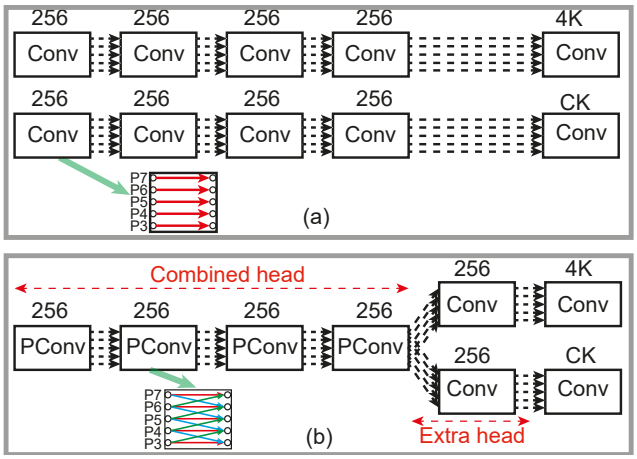

Figure 5: (a) Head design of the original RetinaNet; (b) Head design with PConv. In the final output convolution, K is the number of anchor boxes, which is 1 for anchor-free methods, and C is the number of classes in classification.

4. Scale-equalizing pyramid convolution

In designing the pyramid convolution, we have used a naive implementation. The kernel size keeps constant when the kernel strides along the scale dimension, even though the feature map size shrinks. This is reasonable when PConv is conducted on a Gaussian pyramid (a Gaussian pyramid is built by consecutively Gaussian blurring an image followed by a downsampling) since

Remark 1 *Pyramid convolution is able to extract scale-invariant features from a Gaussian pyramid.*

The detailed mathematical proof can be found in Appen. 2. It is shown intuitively in Fig. 6a. When a shared 2-D kernel (also viewed as a PConv with a scale kernel size of 1) extracts features from the pyramid, objects of different scales can be captured by the same kernel at different level. The Gaussian blur is also necessary in generating the pyramid so as to avoid high-frequency noises in extracting features in downsampled images. On the other hand, too strong blur conceals details. The optimal blurring kernel in the Gaussian pyramid is around the size of the downsampling ratio between two pyramid levels.

In the naive implementation of PConv, and also in the design of RetinaNet head, such a fashion is directly used to process feature pyramid. However, the optimal blurring kernel is hardly satisfied for feature pyramid. In Fig. 6b, we see that as we go to high-level features, the blurring effect of feature maps becomes much more serious than that in an image pyramid. This is due to the many layers of convolution and non-linearity operations in the backbone between two feature maps in the feature pyramid.

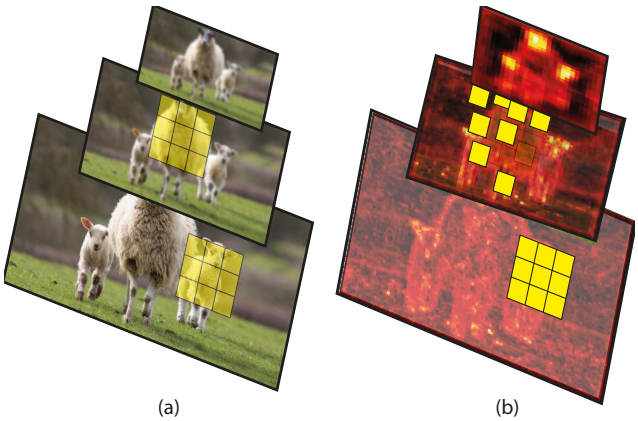

Figure 6: (a) PConv on Gaussian pyramid; (b) SEPC on Feature pyramid

In order to compromise the stronger blurring effect and extract scale-invariant features, some studies advocated using dilated convolution [39]. That is, as the PConv module strides in the scale dimension, the kernel should also be larger than the one used in the bottom-most features. However, because of the non-linearity operations in the backbone, the dilation ratios of different pixels are also different, making it difficult to directly use a constant one.

Instead, we borrow the idea of deformable convolution to directly predict the offset of the convolutional kernel as pconv strides upward in the scale dimension. Shown as Fig. 6b, the kernel of the PConv at the bottom feature map is fixed as a normal 3×3 convolution. As it processes high-level feature maps in the feature pyramid, a deformation offset is predicted based on the current layer of feature map.

There are multiple benefits of this design. 1) The larger blurring effect between two layers of feature pyramid is considered due to its dilating ability of a deformable convolution kernel; 2) Since the computational cost of a convolution reduces by 4 from one layer to its upper feature pyramid level, adding deformable convolution to the high-level feature maps incurs minimal computations. In this study, we study the effect of both SEPC-full that applies SEPC to both the combined and the extra head in Fig. 5b, and SEPC-lite that applies SEPC to only the extra head.

5. Experiments

The experiments in this study were performed on the MS-COCO2017 detection dataset [19] of 80 categories. The training set is composed of around 118k images, and the validation set consists of 5k images (`minival`). The detection metrics are reported by default on `minival`. The results on the test set (`test-dev`) with inaccessible labels are also reported for several models in this study. More de-

tails in regards to the training & inference and speed test settings can be found in Appen. 4 & 6.

5.1. Single-stage object detectors

The single stage object detectors in this study are mostly the latest and state-of-the art models, using either anchor-based such as RetinaNet, [18] and FreeAnchor[43] or anchor-free methods such as FSAF[45] and Reppoints [41]. The results between the proposed pconv and their original baselines are compared in Tab. 1. SEPC-full is found able to boosts the performance by more than 4AP, yet incurring unnecessary inference time increase due to the multiple deformable operations at PConv modules. The improvement of SEPC-lite in each network is also substantial, increasing $3.1 \sim 3.8$AP with only 7% latency increase. It should be noted that our own implemented baseline of FSAF (see details in Append. 5) already achieves 36.9AP, 1.1AP higher than the original results. And the performance of FSAF is further boosted to 40.9 by SEPC-lite, which is even 0.5AP higher than that of Cascade and Deformable Faster-RCNN while maintaining more than 20% faster. The improvement of SEPC-lite on FreeAnchor, one of the best single-stage detectors, is also surprising, making it achieve 41.7 and painlessly furthering the state of the art by 3.2.

5.1.1 Ablation study

In order to figure out the individual performance of pyramid convolution, integrated BN and scale-equalizing head design, we ran a series of ablation studies.

5.1.2 Effect of each component

The replacement of normal convolution in the head with PConv brings around $\sim 1.5AP$ increase in various models. As for the speed of PConv, the total FLOPs of pconv is actually smaller than the original head, yet the latency still increases by around 3%. This small increased computational cost is partially because implementation of pyramid convolution is approximated by 3 independent kernels in each layer, instead of a direct 3-D convolution which can fully leverage parallel computations.

The insertion of integrated BN (iBN) in the head also benefits the model by $0.2 \sim 1.2$AP for different architectures. Several other studies also advocated group normalization (GN) when training detection networks[35]. However, one trait of BN that is missing in GN is that BN does not require calculating on-site statistics when conducting inference and can be merged in the former convolutional layer. This brings significant advantage in inference speed, as revealed by the same forward latency with iBN. The increased performance is a natural result of the faster optimization and better generalization of BN. The fast convergence of the model with iBN can be found in Appen. 3.

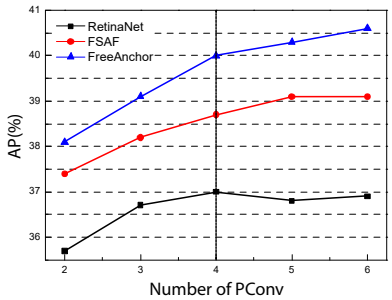

Figure 7: AP change with number of pyramid convolution in different architectures.

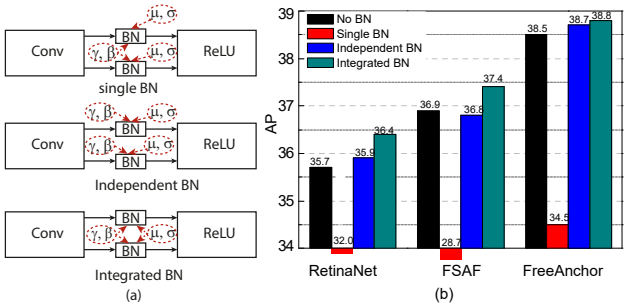

Figure 8: (a) Different batch normalization implementations in the feature pyramid levels. Only 2 feature levels are given as an illustration; (b) Comparison of AP results of architectures with different batch normalization implementations in the feature pyramid.

As for the effect of scale-equalizing head, we also compare the results of SEPC with that only without scale-equalizing head and find significant improvement (~ 1.5AP increase), showing further potential applications if deformable convolution operation can be accelerated.

5.1.3 Effect of the number of pyramid convolution stacks

The total number of pyramid convolutions is adjusted from 2 to 6. The recorded AP of different detectors is shown in Fig. 7. The figure illustrates that all these three detectors benefit from increasing the number of PConv from 2 to 4 due to the gradual information flow from top to bottom by PConv stacks. Using four stacked PConv in heads is rational as it provides descent average precision without causing much redundancy.

Detector	Note	FLOPS(G)	Time(ms)	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RetinaNet	baseline	239.32	73.2	35.7	55.0	38.5	18.9	38.9	46.3
	pconv	239.29	76.5	37.0	57.7	39.4	22.3	40.8	48.9
	pconv+iBN	239.36	76.4	37.9	59.3	40.6	22.5	42.2	49.1
	SEPC	250.16	89.6	**39.7**	**60.4**	**42.7**	**23.1**	**44**	**52.2**
	SEPC-lite	239.9	78.5	38.8	59.9	41.8	22.6	42.8	51
FSAF	baseline*	205.2	62.4	36.9	56.1	39	20.6	40.1	48.2
	pconv	205.18	66.0	38.7	58.9	41.1	22.2	42	51
	pconv+iBN	205.25	66.1	38.9	59.1	41.8	22.2	42.5	51
	SEPC	216.05	77.4	**41.3**	**60.4**	**43.6**	**23**	**44.8**	**57.8**
	SEPC-lite	205.79	68.2	40.7	60	43.4	22.4	44.6	55.1
FreeAnchor	baseline	239.32	76.4	38.5	57.3	41.2	21.1	41.8	51.5
	pconv	239.29	79.4	40.0	59.1	43	22.8	43.8	53.3
	pconv+iBN	239.36	79.7	41.2	60.5	44.3	24.3	44.6	54.6
	SEPC	250.16	89.9	**42.8**	**61.9**	**45.9**	**25.6**	**46.4**	**57.4**
	SEPC-lite	239.90	81.2	41.7	61	45.1	24.2	45.2	54.8

*: our own implementation

Table 1: Comparison of detection AP results of different architectures. All models were trained using ResNet-50 backbone and adopted the `1x` training strategy. Results were evaluated on COCO `minival` set.

5.1.4 Comparison of different BN implementations in the head

There are different implementations of BN in dealing with feature pyramids, which are shown in Fig. 8a. The output after each BN module is $y = \gamma\frac{x-\mu}{\sigma} + \beta$, where γ and β are parameters, and μ and σ are batch statistics for normalization. Single BN adds a BN module after each feature pyramid level with shared parameters γ and β across the pyramid. But each feature map collects statistics on its own during training. Independent BN makes both parameters and statistics of BN in a feature layer exclusive to itself and is used in [7]. Integrated BN, as discussed in section 3.2, calculates batch statistics over all feature maps from feature pyramid networks.

Effect of these four BN designs are demonstrated in Fig. 8. Trivially using single BN results in a catastrophic decline in performance of detectors, due to the mismatch between the shared parameters and non-shared statistics. Integrated BN and independent BN both improves AP, and integrated BN outperforms independent BN because of the more stable statistics during training.

5.1.5 Comparison with other feature fusion modules

In regards to different feature fusion methods, Tab. 2 presents our comparison of PConv to other state-of-the-art feature fusion modules on FreeAnchor. It is obvious that PConv provides a dramatic performance increase compared to common feature pyramid networks, including NAS-FPN[7] and Libra[27]. Moreover, the designed PConv stack head also earns the minimum FLOPs increase among the feature fusion modules. Results of this section validate the effectiveness of PConv in feature fusion.

Feature fusion	AP	AP_{50}	AP_{75}	FLOPS(G)
FPN	38.5	57.3	41.2	239.3
HRNet	38.6	57.1	41.3	297.58
PANet	38.9	57.6	41.6	245.9
NAS-FPN	39.1	57.0	41.8	347.1
Libra	39.4	58.7	42.2	315.8
PConv	40.0	59.1	43	239.2

Table 2: Comparison of Pconv with other feature fusion modules on FreeAnchor. Results evaluated on COCO `minival` are reported.

5.2. Comparison with state-of-the-art object detectors

In this section, we compare our method to other state-of-the-art object detectors on COCO2017 benchmarking dataset. The training strategies followed `2x` with 640-800 scale jitter and the results were obtained with only a single scale. Details can be found in Appen. 4. We only report FSAF and FreeAnchor equipped with SEPC-lite for the purpose of real potential applications since SEPC-lite incurs intangible computation cost for large backbones such as ResNet-101 and ResNext-101. It is observed that SEPC-lite boosts the original baselines by a significant margin and achieves the state-of-the-art 47.1AP without utilizing bells and whistles (e.g. multi-scale test, sync BN, deformable backbone), surpassing even most two-stage detectors with deformable backbones and multi-scale test.

Method	Backbone	Iter	Input Size	AP	AP_{50}	AP75	AP_s	AP_m	AP_l
Two-Stage Detectors									
Faster-RCNN^{-}[29]	ResNet-101	180k	min800	39.4	60.6	43.0	22.1	43.6	52.1
Cascade-RCNN^{-}[1]	ResNet-101	150k	min800	42.5	60.7	46.3	23.7	46.1	56.9
TridentDet [15]	ResNet-101	270k	min800	42.7	63.6	46.5	23.9	46.6	56.6
SNIP* [31]	ResNet-101-Deformable	-	-	44.4	66.2	49.9	27.3	46.4	56.9
SNIPPER* [32]	ResNet-101-Deformable	-	-	46.1	67.0	51.6	29.6	48.9	58.1
TridentDet$^{*\square\triangle}$ [15]	ResNet-101-Deformable	270k	min800	46.8	67.6	51.5	28.0	51.2	60.5
Faster-RCNN^{-}[8]	ResNext-101-64d	180k	min800	40.7	62.0	44.6	22.9	44.5	53.6
Cascade-RCNN^{-}[1]	ResNext-101-64x4d	150k	min800	44.7	63.1	49.0	25.8	48.3	58.8
One-Stage Detectors									
AlignDet[3]	ResNet-101	135k	min800	42.0	62.4	46.5	24.6	44.8	53.3
FreeAnchor[43]	ResNet-101	180k	min800	42.6	61.3	46.2	25.3	46.4	56.0
FSAF[45]	ResNet-101	180k	min800	41.2	62.4	45.8	25.6	46.8	54.9
RetinaNet[18]	ResNext-101-64x4d	180k	min800	39.6	60.3	42.3	21.6	43.5	53.5
FSAF[45]	ResNext-101-64X4d	180k	min800	43.6	63.7	46.7	26.1	48.0	56.5
FreeAnchor[43]	ResNext-101-64x4d	180k	min800	44.4	63.8	48.0	27.3	48.5	57.8
AlignDet[3]	ResNext-101-32x8d	135k	min800	44.1	64.7	48.9	26.9	47.0	54.7
CornerNet[14]	Hourglass-104	500k	511	40.6	56.4	43.2	19.1	42.8	54.3
CenterNet[5]	Hourglass-104	480k	511	44.9	62.4	48.1	25.6	47.4	57.4
FSAF+SEPC-lite	ResNet-101	180k	min800	**44.8**	64.5	48.1	25.4	48.5	58.6
FreeAnchor+SEPC-lite	ResNet-101	180k	min800	**45.5**	64.9	49.5	27	48.8	56.7
FreeAnchor+SEPC-lite	ResNext-101-64x4d	180k	min800	**47.1**	67.0	51.2	29.3	50.8	58.3

*: Multi-scale testing; $^{-}$: single-scale training; $^{\square}$: soft-nms applied; $^{\triangle}$: synchronized BN

Table 3: Comparing our SEPC-lite with other state-of-the-art object detectors. Results evaluated on `test-dev`. Methods adopting significantly longer training time or much larger input sizes are not included in the table.

Detector	Note	AP	AP_{50}	AP_{75}
Faster	Baseline	36.5	58.4	39.1
	Pconv	38.5	59.9	41.4
Mask	Baseline	37.3	59	40.2
	Pconv	39.6	60.1	43.5
HTC	Baseline	42.1	60.8	45.9
	Pconv	43.6	62.0	47.4

Table 4: Extension of only PConv module to two-stage detectors including Faster R-CNN[29], Mask R-CNN[10] and HTC[2].

5.3. Extension to two-stage object detectors

So far our discussion mainly focuses on single-stage object detectors, in this section, we present that PConv can still be an effective module when applied to two-stage object detectors. As shown in Tab 4, PConv provides remarkable improvement of AP on different two-stage detectors. PConv provides the most AP increase to Mask-RCNN, which improves the AP by 2.3.

Conclusion

In this study, we explore considering the inter-scale correlation through a pyramid convolution (PConv), which runs 3-D convolution on both the scale and spatial dimension of the feature pyramid. However, the striding pattern for this PConv in both spatial and scale dimension is quite different from conventional ones. First, due to the different spatial sizes in the pyramid, the stride number of the spatial slices of PConv kernels is proportional to the convolved feature map size in the pyramid level. This pendulum-alike striding pattern of the PConv kernel helps align the spatial position of neighboring feature maps as they are involved in one PConv. Second, when PConv strides up in the scale dimension, the kernel should also adjust its spatial deformation as well, which is then called scale-equalizing pyramid convolution (SEPC). The naive striding pattern with a fixed spatial kernel size actually best suites for extracting features in a Gaussian pyramid, which is quite far from the feature pyramid generated by deep networks. And SEPC helps relax this discrepancy and extract more robust features. Being light-weighted and compatible with most single-stage detectors, SEPC is able to significant improve the detection performance with minimal computational cost increase.

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